POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE | Full Marks : 100
Course: Machine Learning (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the major issues and challenges in machine learning such as overfitting, underfitting, data quality, and computational cost. 7

   b) What do you mean by regularization techniques? Explain the different types of regularization techniques with necessary derivations. 8

2. a) Construct a Decision Tree using the ID3 algorithm for the following dataset. 8

Student	Study	Attendance	Pass
A	Yes	High	Yes
B	Yes	Low	No
C	No	High	Yes
D	No	Low	No
E	Yes	High	Yes
F	No	High	Yes

   Calculate entropy and information gain at each step.

   b) Explain SVM with its type, kernel, applications and mathematics involved. 7

3. a) Given the points A(3, 7), B(4, 6), C(5, 5), D(6, 4), E(7, 3), F(6, 2), G(7, 2) and H(8, 4), Find the core points and outliers using DBSCAN. Take Eps = 2.5 and MinPts = 3. 8

OR

What are the different type of clustering? Consider the one-dimensional data points 18, 22, 25, 27, 42, and 43 and apply agglomerative clustering to cluster and plot the Dendrogram.

   b) Describe the role association rule mining. Given the following transaction set find the frequent itemset using Apriori Algorithm. 7

Assume Minimum support item set using support=2, confidence = 75%. Also, evaluate the association rules

TID	Items
1	1, 3, 4
2	2, 3, 5
3	1, 2, 3, 5
4	2, 5

4. a) Describe the Multilayer Perceptron (MLP) architecture. Explain the backpropagation algorithm used for training neural networks. 7

OR

A single layer perceptron has inputs x1 = 1, x2 = 0.5 with weights w1 = 0.4, w2 = 0.6, bias b = -0.3, and learning rate $\eta = 0.1$. If the target output is 1 and the perceptron produces output 0 using a step activation function, show one iteration of weight updates using the perceptron learning rule.

   b) What is vanishing and exploding gradient problem in RNN? How LSTM resolves these problem? Explain. 8

5. a) Explain CNN architecture? Explain its architecture and how does CNN extract features from the image? Describe with suitable example. 8

   b) What are the roles of loss function in neural network? Describe the classification loss functions. 7

6. a) Explain the concept of training, validation and testing data. How does K-fold cross validation work? 7

   b) What is confusion matrix? Why it is important? Based on the given confusion matrix, calculate accuracy, precision, recall and f1 score. 8

		Prediction: Yes	Prediction: No
Actual	Yes	40 TP	10 FN
	No	15 FP	60 TN

7. Write short notes on: **(Any two)** 2×5

   a) Machine learning workflow

   b) PCA

   c) Regression metrics

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